

Management of Male Breast Cancer: ASCO Guideline Summary

Michael J. Hassett, MD, MPH¹; Mark R. Somerfield, PhD²; and Sharon H. Giordano, MD, MPH³

Male breast cancer is a rare disease representing < 1% of all breast carcinomas diagnosed in the United States. In 2019, an estimated 2,670 new cases of breast cancer will be diagnosed among men in the United States, and about 500 men will die of breast cancer.¹ The lifetime risk of breast cancer is about 1 in 1,000 for a man, whereas it is approximately 1 in 8 for a woman. Breast cancer incidence rates increase steadily with age in men as they do in women; however, the average age of a new breast cancer diagnosis is 5 years older for men (67 years) than for women (62 years). Other risk factors for male breast cancer include a family history of breast cancer, black race, exposure to radiation to the breast or chest, carrying a predisposition germline genetic mutation (eg, *BRCA2*, *BRCA1*, *CHEK2*, *PALB2*), use of exogenous estrogen, and diseases associated with hyperestrogenism (eg, Klinefelter syndrome).²

There are substantial knowledge gaps concerning the optimal management of breast cancer in men. To date, approaches to treating men with breast cancer have been extrapolated largely from research conducted in women with breast cancer. Planned or ongoing treatment trials that focus on men should eventually inform the standard of care, but they are years from completion and will not address all important questions for male breast cancer. The ASCO guideline offers recommendations on several key aspects of the management of male breast cancer.³ Additional information is available

at www.asco.org/breast-cancer-guidelines. Patient information is available at www.cancer.net.

WHAT IS PRACTICE CHANGING?

The Panel identified aspects of management for which practice among men and women should be similar. These include the approaches to the use of gene expression profile testing to guide adjuvant treatment decision making; the use of primary surgery and adjuvant radiation therapy (acknowledging that lumpectomy followed by radiation is an acceptable option for men, even though it may be used less frequently than for women); the use of chemotherapy in the adjuvant and advanced/metastatic disease settings; the use of targeted therapies in the advanced/metastatic settings (excluding endocrine therapy); and the management of treatment-related adverse effects. The Panel also identified aspects of management for which practice among men and women differed. These included the approaches to the use of endocrine therapy in the adjuvant and metastatic settings, the use of genetic testing for inherited risk factors, and survivorship care. Tamoxifen is recommended as first-line endocrine therapy for men in the adjuvant and advanced/metastatic settings. Testing for inherited predisposing risk factors is recommended routinely for men. Consensus regarding survivorship care for men is limited, but ipsilateral surveillance mammography for men treated with lumpectomy is recommended routinely.

Author affiliations and support information (if applicable) appear at the end of this article.

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THE BOTTOM LINE

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Guideline Question

What is the optimal management for men with breast cancer including use of adjuvant endocrine therapy, use of endocrine therapy for advanced or metastatic disease, targeted therapies, management of treatment-related adverse effects, genetic testing, and posttreatment surveillance?

Target Population

Men diagnosed with invasive breast cancer.

Target Audience

Medical oncologists, radiologists, radiation oncologists, surgical oncologists, endocrinologists, oncology nurses, patients/caregivers/advocates, oncology advanced practice providers, genetic counselors.

Methods

An Expert Panel was convened to develop clinical practice guideline recommendations based on a systematic review of the medical literature. Guideline development involved a formal consensus process.

Results of Phase I: The Breast Cancer Advisory Group Guideline Survey

The Advisory Group reported high-level consensus that the following components of management were largely the same for men and women: use of gene expression profile testing to guide adjuvant treatment decision making (eg, *Oncotype Dx* [Genomic Health, Redwood City, CA] and prognostic tests), primary surgery, adjuvant chemotherapy, adjuvant radiation therapy, and chemotherapy for advanced/metastatic disease.

Clinical implications. Based on these findings, the Expert Panel recommends that, for men with breast cancer, the approach to the use of gene expression profile testing to guide adjuvant treatment decision making; to primary surgery; to adjuvant chemotherapy; to adjuvant radiation therapy; and to chemotherapy for advanced/metastatic disease should be the same as the approach used for women.

Results of Phase II: The Expert Panel Consensus Development Process

The consensus development process focused on topics for which the Advisory Group and Expert Panel did not believe there was already widespread agreement that management of men and women was largely the same: endocrine therapy in the adjuvant setting, endocrine therapy in the metastatic setting, management of adverse effects from endocrine therapy, germline genetic testing, survivorship care, and the use of targeted therapies for advanced/metastatic disease. Over 2 rounds of ratings, 12 recommendations pertaining to 10 clinical questions derived from these 6 topics satisfied the required 75% consensus agreement threshold.

Recommendations

Recommendation 1.1. Men with hormone receptor–positive breast cancer who are candidates for adjuvant endocrine therapy should be offered tamoxifen (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Recommendation 1.2. Men with hormone receptor–positive breast cancer who are candidates for adjuvant endocrine therapy but have a contraindication to tamoxifen may be offered a gonadotropin-releasing hormone (GnRH) agonist/antagonist and an aromatase inhibitor (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 2.1. Men who are treated with adjuvant endocrine therapy should be treated for an initial duration of 5 years (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Recommendation 2.2. Men who have completed 5 years of tamoxifen, have tolerated therapy, and still have a high risk of recurrence may be offered an additional 5 years of tamoxifen therapy (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Recommendation 3. Men with early-stage breast cancer should not be treated with bone-modifying agents to prevent recurrence but could still receive these agents to prevent or treat osteoporosis (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 4.1. Men with advanced or metastatic, hormone receptor–positive, HER2-negative breast cancer should be offered endocrine therapy as first-line therapy except in cases of visceral crisis or rapidly progressive disease.

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THE BOTTOM LINE (CONTINUED)

Options include tamoxifen, an aromatase inhibitor with a GnRH agent, and fulvestrant. CDK4/6 inhibitors can be used in men as they are used in women (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Recommendation 4.2. Men who develop recurrent metastatic, hormone receptor–positive, HER2-negative breast cancer while receiving adjuvant endocrine therapy should be offered an alternate endocrine therapy except in cases of visceral crisis or rapidly progressive disease (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Recommendation 4.3. Endocrine therapy for men with advanced or metastatic, hormone receptor–positive, HER2-negative breast cancer may be sequenced as in women (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 5. Targeted therapy guided by HER2, PDL-1, PIK3CA, and germline *BRCA* mutation status may be used in the treatment of advanced or metastatic male breast cancer using the same indications and combinations that are offered to women (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong). [Targeted therapy based on hormone receptor status is addressed in Recommendations 4.1-4.3.]

Recommendation 6.1. Management of endocrine therapy toxicity is similar to the approach used for women (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 6.2. Testosterone/androgen supplementation should not be used by men with breast cancer (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 7. Physicians should counsel patients about the symptoms of recurrence, including new lumps, bone pain, chest pain, dyspnea, abdominal pain, or persistent headaches. The risk of breast cancer recurrence continues through 15 years after primary treatment and beyond. Continuity of care for patients with breast cancer is recommended and should be performed by a physician experienced in the surveillance of patients with cancer and in breast examination, including the examination of irradiated breasts (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 8. Ipsilateral annual mammogram should be offered to men with a history of breast cancer treated with lumpectomy, if technically feasible, regardless of genetic predisposition (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Recommendation 9.1. Contralateral annual mammogram may be offered to men with a history of breast cancer and a genetic predisposing mutation (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 9.2. Breast magnetic resonance imaging is not recommended routinely in men with a history of breast cancer (Type: formal consensus; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 10. Male patients with breast cancer should be offered genetic counseling and genetic testing for germline mutations (Type: formal consensus; Evidence quality: low; Strength of recommendation: strong).

Draft Recommendations That Did Not Reach the Required 75% Consensus Threshold

Two draft recommendations did not exceed the 75% consensus agreement threshold required for adoption into the formal guideline statement: (1) contralateral annual mammography in men without a genetic predisposing mutation and (2) chemoprevention.

Additional Resources

More information, including a supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/breast-cancer-guidelines. Patient information is available at www.cancer.net.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

AFFILIATIONS

¹Dana-Farber Cancer Institute, Boston, MA

²American Society of Clinical Oncology, Alexandria, VA

³The University of Texas MD Anderson Cancer Center, Houston, TX

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Road, Suite 800, Alexandria, VA 22314; e-mail: guidelines@asco.org.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST AND DATA AVAILABILITY STATEMENT

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AUTHOR CONTRIBUTIONS

Conception and design: All authors

Administrative support: Mark R. Somerfield

Collection and assembly of data: Mark R. Somerfield

Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

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REFERENCES

1. Siegel RL, Miller KD, Jemal A: Cancer statistics, 2019. *CA Cancer J Clin* 69:7-34, 2019
2. Giordano SH: Breast cancer in men. *N Engl J Med* 378:2311-2320, 2018
3. Hassett MJ, Somerfield MR, Baker ER, et al: Management of male breast cancer: ASCO guideline. *J Clin Oncol* doi:[10.1200/jco.19.03120](https://doi.org/10.1200/jco.19.03120)



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